Expanding balloon

The radius r cm of a spherical balloon is increasing at 0.3cm/s.

Find the rate at which the volume is increasing to 3dp when

r=8cm.

$$V = \frac{4}{3}\pi r^3$$

Conical pile

A conical pile of sand always has height 3 times its radius. The radius is increasing at 0.2m / min. Find the rate at which its volume is increasing when r = 5m.

$$V = \frac{1}{3}\pi r^2 h$$

Changing rectangle

The length of a rectangular display screen is increasing at 2 cm/s, while its width is decreasing at 0.5 cm/s.

Find the rate of change of its area when the length is 20 cm and the width is 12 cm. State whether the area is increasing or decreasing.

Inverted Conical Tank

An inverted conical tank has height 10 m and top radius 4 m. Water drains from the bottom of the tank at a constant rate of 2 m3/min. Find the rate at which the water depth is decreasing when h=5 m.

$$V = \frac{1}{3}\pi r^2 h$$

Triangular Water Tank

A water tank is 8 m long. Each end is an isosceles triangle of height 2 m and top width 4 m. Water is pumped into the trough at 0.6 m3 /min. Find the rate at which the water depth is increasing when the depth is 1.5 m.

Surface area of right-angle triangle with two equal lengths x: $\frac{1}{2}x^2$

Fixed volume open box

A open box (no top) with a square base must hold 500cm3.

What is the surface area of the square base to minimize the amount of material required to make it?

Poster Design

A poster must contain 600 cm2 of printed material. It has margins of 3 cm on the left and right and 4 cm at the top and bottom. Find the minimal surface area of the poster.

Water Flowing Into Tank

Water enters a tank at a rate of 4t+6 liters per minute, where t is the number of minutes after filling begins. Initially, the tank contained 20 liters of water. Find the volume of water after 5 minutes.

Distance travelled

The velocity of a car at time t is given by

$$v(t) = 3t^2 - 12t + 9$$

What's the total distance travelled by the car after 5 seconds?

Daily profit

The daily revenue rate, in dollars per day, of a company at time *t* days is modeled as:

$$R(t) = 8t - t^2$$

The daily cost rate is modeled as:

$$C(t) = 2t + 3$$

What is the total profit of the company over the first five days, assuming the functions are continuous?